Girayhan Sultan Çakıroğlu

Istanbul, Türkiye | girayhancakiroglu7@gmail.com | giraycakr.github.io | github.com/giraycakr

Research Interests

Machine learning and deep learning; trustworthy AI, including interpretability, robustness and shortcut learning; natural language processing and large language models.

Education

Istanbul Medeniyet University Istanbul, Türkiye
B.Sc. in Computer Engineering *Sep 2024 – Jun 2027*

GPA: 3.37/4.00 (2025–26 academic year: 3.67/4.00)

Selected coursework: Natural Language Processing (4.0), Introduction to Data Science (4.0), Probability and Statistics (4.0), Discrete Mathematics (4.0), Deep Learning (3.5), Pattern Recognition (3.5)

Marmara University Istanbul, Türkiye
Mechatronics Engineering (coursework; transferred to Istanbul Medeniyet University) *Sep 2022 – Jun 2024*

Publications

[1] **Girayhan Sultan Çakıroğlu** and Ertürk Erdağı. A Multi-Metric Explainable Artificial Intelligence Analysis of Data Augmentation Methods on CIFAR-10 and CIFAR-100. Under review at *Istanbul Commerce University Journal of Science*, 2026.

Research Experience

Undergraduate Researcher, Istanbul Medeniyet University Sep 2025 – Jul 2026
Advisor: Asst. Prof. Ertürk Erdağı

- Designed a controlled experimental study of six data augmentation methods and a baseline on CIFAR-10/100 with ResNet-18 (42 training runs: 7 configurations $\times$ 2 datasets $\times$ 3 seeds), evaluated with four interpretability techniques: multi-layer Grad-CAM, attention coverage, feature-space silhouette analysis and occlusion sensitivity.
- Showed that the commonly reported link between class difficulty and augmentation gain is a ceiling-effect artifact: the raw correlation ($r = -0.83$) disappears after normalizing for the remaining accuracy gap ($r = +0.04$), with CutMix closing on average about 45% of that gap, independent of class difficulty.
- Found that CutMix narrows the model's attention onto the object rather than spreading it (coverage -18% on CIFAR-100), while qualitative occlusion maps show its decisions drawing on several object regions and less on background shortcut cues.
- Showed that no single interpretability metric explains per-class accuracy gains, arguing for multi-metric evaluation of augmentation methods.
- Developed the work from my senior capstone project into a first-author manuscript [1]; built the PyTorch pipeline, produced all figures and tables, and wrote the original draft.

Projects

Privacy-First Legal Document Analysis Platform 2025
Natural Language Processing (BIL482), individual course project

- Built a fully local retrieval-augmented generation (RAG) pipeline with LangChain, Ollama and ChromaDB, so that confidential contracts never leave the user's machine.
- Added a Streamlit interface with automated risk scoring and persona-based answers.

Industry Experience

School Bus Tracking and Route Planning Platform, *Sole Developer* 2026 – Present

- Built the platform alone; licensed annually to two school transportation companies, in daily use.
- Built an AI assistant that answers staff questions from each company's own operational data and shows the source records behind each answer (RAG with a locally hosted LLM, LangChain and ChromaDB).
- Applied ML in the platform's accounting module to flag anomalous payment records for review.
- FastAPI/PostgreSQL backend, two React web consoles, two React Native apps; real-time tracking.
- Solved route planning as a capacitated vehicle routing problem using OR-Tools and OSRM.

Turkcell, *Service Reliability Engineering Intern* Aug 2025 – Sep 2025

- Worked on machine learning and service reliability tasks at one of Türkiye's largest telecom companies.
- Examined Jenkins CI/CD pipelines; found and reported a UI defect on the live customer-facing website.

Yıldız Holding, *Enterprise Solutions Intern* Jun 2025 – Jul 2025

- Supported SAP SD/MM sales, procurement and inventory workflows at the owner of Godiva and Ülker.

Atlantis Otomasyon, *Software Engineering Intern* Jun 2024 – Sep 2024

- Built a full-stack Python application for meteorological data collection and reporting.
- Added threshold-based and statistical outlier detection that discarded or flagged faulty sensor readings.

Extracurricular Activities

Google Developer Student Clubs, Marmara University, *Vice President* Nov 2022 – Aug 2023

- Co-organized technical visits to six companies and two workshops that together reached hundreds of students.
- Launched "Tech Talk on the Ferry," a Bosphorus ferry event where engineers and industry leaders give talks to students on board; the club still runs it.

Çorbada Tuzun Olsun Association, *Volunteer* Nov 2022 – Aug 2023

- Distributed meals to people experiencing homelessness in Istanbul.

TEMA Foundation, *Volunteer* Mar 2024 – Apr 2025

- Took part in tree planting and coastal clean-ups with Türkiye's leading environmental foundation.

Test Scores

TOEFL iBT (Oct 2026): 5.0/6.0 overall (100/120, CEFR C1)

Reading 5.0 | Listening 6.0 | Writing 5.0 | Speaking 4.0

Skills

Programming: Python, SQL, TypeScript, C

Machine Learning and NLP: PyTorch, scikit-learn, NumPy, Pandas, LangChain, Ollama, ChromaDB

Tools: FastAPI, PostgreSQL, React, React Native, Docker, Git, LaTeX

Languages: Turkish (native), English (professional proficiency)